TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Conversion of 98% H_2SO_4 and 37% HCl into Suitable Concentration

From percentage and density to molarity — the master formula, the two bench acids computed, and the dilution arithmetic and safety of preparing working solutions.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 12 · LEVEL 1 TERM 1

1. Aim and the data on the bottle

Aim: to convert the concentrated commercial acids into solutions of a suitable (chosen) concentration. A reagent bottle states its strength as a **mass percentage** and a **density** (or specific gravity): concentrated sulphuric acid ≈ 98% w/w, density 1.84 g mL^{-1}; concentrated hydrochloric acid ≈ 37% w/w, density 1.19 g mL^{-1}. Percentage and density between them hide the molarity, and one short derivation recovers it.

2. The master formula

Take 1 litre (1000 mL) of the acid. Its mass is 1000 × d grams (d = density in g mL^{-1}). Of that mass, the pure acid is (percentage/100) × 1000 × d grams. Dividing by the molar mass M gives moles per litre:

$$\text{Molarity} = (\%\ \text{w/w} \times d \times 10) / M$$

ACID	SUBSTITUTION	MOLARITY OF THE CONCENTRATE
H_2SO_4 (98%, d = 1.84, M = 98)	(98 × 1.84 × 10) / 98	**18.4 M** (≈ 36.8 N, dibasic)
HCl (37%, d = 1.19, M = 36.5)	(37 × 1.19 × 10) / 36.5	**≈ 12.1 M** (= 12.1 N, monobasic)

These two numbers — 18.4 and 12 — are worth memorising; half the dilution arithmetic of the laboratory starts from them.

3. Dilution to the required concentration

Moles are conserved (Experiment 11): $M_1V_1 = M_2V_2$, with 1 = concentrate, 2 = target.

WORKED EXAMPLES

(a) 500 mL of 0.5 M H_2SO_4 from the concentrate: $V_1 = M_2V_2/M_1 = 0.5 \times 500 / 18.4 =$ **13.6 mL** of concentrated acid, diluted to 500 mL. **(b) 250 mL of 1.0 M HCl:** $V_1 = 1.0 \times 250 / 12.1 =$ **20.7 mL**, diluted to 250 mL. **(c) 1 litre of 0.1 N H_2SO_4:** 0.1 N = 0.05 M; $V_1 = 0.05 \times 1000 / 18.4 =$ **2.72 mL** — small volumes like this are measured by burette or graduated pipette, never guessed.

4. Procedure and safety — the acid-into-water law

1. Calculate the concentrate volume (above); write the calculation in the record before touching the bottle.
2. Place about two-thirds of the final volume of **distilled water** in the flask/beaker first.
3. Add the acid **slowly, in a thin stream, with stirring** — always **acid into water, never water into acid**: dilution of H_2SO_4 is fiercely exothermic, and water landing on concentrated acid boils on contact and spits acid. (The school mnemonic: “Do as you oughta — add acid to water.”)
4. For H_2SO_4 especially, let the mixture **cool to room temperature** before making up to the mark — a hot solution contracts as it cools and the volume (hence concentration) would be wrong.
5. Make up to the mark, mix by repeated inversion, label with concentration, date and preparer. HCl solutions are best standardised soon after (Experiment 15), since the concentrate’s percentage varies from bottle to bottle around 37%.

5. Precautions

- Concentrated acids are handled with the bottle over a tray, goggles on, and small quantities only; spills are flooded with water and reported.
- Never pipette by mouth — a filler, always.
- Check the bottle’s actual percentage and density: the formula uses *this* bottle’s numbers; “about 37%” stock varies, and exact work standardises afterwards regardless.
- Store the diluted acid labelled and stoppered; HCl fumes attack neighbouring reagent labels and metal fittings.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Derive the molarity of a 69% w/w HNO_3 of density 1.42 g mL^{-1} (M = 63), using the master formula.
2. Calculate the volume of concentrated H_2SO_4 (18.4 M) needed for 1 litre of 2.0 M acid.
3. Calculate the volume of concentrated HCl (12.1 M) needed for 100 mL of 0.5 M acid.

4. Why is acid added to water and not the reverse, stated in terms of heat?
5. Why must a freshly diluted H_2SO_4 solution cool before it is made up to the mark?
6. The bottle says 37% but the true strength of an old, often-opened bottle is lower. Which later experiment repairs the uncertainty, and how?

Answer key

1. M = (69 × 1.42 × 10)/63 = 979.8/63 ≈ 15.6 M.
2. V_1 = 2.0 × 1000/18.4 ≈ 108.7 mL.
3. V_1 = 0.5 × 100/12.1 ≈ 4.13 mL.
4. Dilution (of H_2SO_4 above all) releases much heat. Acid added slowly to a large volume of water lets the water absorb and spread the heat; water added to concentrated acid heats at the surface layer, boils violently, and spits concentrated acid outward.
5. Because volume is temperature-dependent: a hot solution made to the mark will contract on cooling below the mark, leaving the concentration higher than intended. Cooling first makes the marked volume true at working temperature.
6. Standardisation (Experiments 13–15): the prepared solution is titrated against a primary standard (for HCl, standard Na_2CO_3), and its exact concentration is computed from the titre — the label then carries the measured, not the assumed, strength.
